



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Spring Semester Examination 2022-23

Date of Examination: April 18, 2023

Session: AN

Duration: 3 hrs. Full Marks: 50

Subject No.: CH61014

Subject Name: Advanced Heat Transfer

Department/Center/School: Chemical Engineering

Specific charts, graph paper, log book etc., required: No

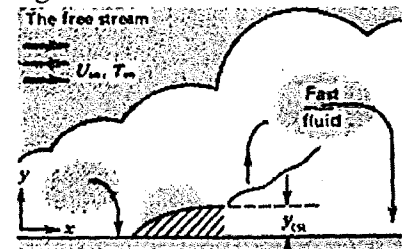
Special Instructions (if any): Write the answers to each question in one place. Assume, if necessary, clearly stating them. No queries will be entertained during the examination.

Part-A

1. (a) Formulate the momentum balance equation with appropriate boundary conditions for a natural convective flow at the boundary layer of a vertical plate kept at a uniform constant temperature T_0 , with the surrounding fluid temperature at T_∞ . Include Boussinesq approximation in your formulation.
- (b) What is the condition for the flow to be dominated by natural convection over forced convection?
- (c) Estimate the order of magnitude of the heat transfer coefficient and Nusselt number for high Pr number fluids for the above system if the flow is dominated by natural convection. [4+3+3=10]
2. A fast-flowing fluid (turbulent flow) with a free stream velocity U_∞ , is flowing over a flat plate, as presented in the figure. The momentum and energy equation in the boundary layer of a turbulent flow can be expressed using the time-averaged velocity and temperature and given as

$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} = -\frac{1}{\rho} \frac{d\bar{P}}{dx} + \frac{\partial}{\partial y} \left[(\nu + \epsilon_M) \frac{\partial \bar{u}}{\partial y} \right] \quad (1)$$

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right] \quad (2)$$



Where, ϵ_M and ϵ_H are empirical functions and known as momentum eddy diffusivity and thermal eddy diffusivity respectively.

Prandtl's mixing length model is used for the velocity distribution near the wall. For a fully turbulent region the relation between momentum diffusivity and momentum eddy diffusivity using Prandtl's mixing length model is given as, $\frac{\epsilon_M}{\nu} = \frac{dy^+}{du^+} = \kappa y^+$; where κ is the empirical constant equal to 0.45;

The turbulent Prandtl number for the flow is $Pr_t = \epsilon_M / \epsilon_H \approx 0.9$ and the dimensionless conduction sublayer (CSL) thickness in wall coordinate is given as $y_{CSL}^+ \approx 20.1$. The relevant variables in the wall coordinate are

$$u^+ = \bar{u} / u_*; \quad v^+ = \bar{v} / u_*; \quad x^+ = xu_* / \nu; \quad y^+ = \frac{yu_*}{\nu}; \quad T^+(x^+, y^+) = (T_0 - \bar{T}) \rho c_p u_* / q_0''$$

All the symbols stand for conventional variables and constants.

- (a) Derive the temperature profile near the plate (from the plate to the turbulent core) using wall coordinates for the given system with relevant assumptions.
- (b) Do you expect a change in the derived temperature profile if the flow properties of the system are altered keeping the fluid same (assume that the fluid properties are constant)? [4+1=5]

3. An experiment is conducted in a laboratory to examine the wall shear stress on an aeroplane wing, maintained at a constant temperature of 25°C ($T_w = 25^\circ\text{C}$). An air stream of uniform velocity ($U_\infty = 10\text{ m/s}$) and temperature ($T_\infty = 10^\circ\text{C}$) at a distance from the wing is blown over it. Roughness present near the leading edge of the wing makes the flow turbulent from $x = 0$. At distance $x (= 1\text{ m})$ from the plate leading edge in the direction of the flow, the wall shear stress τ_w equal to 0.25 N m^{-2} is measured. In the same location, a wire of a diameter of $1\text{ }\mu\text{m}$ is placed 0.3 mm above the wall to measure the fluid mean temperature, which is found to be equal to 22°C .

- With these experimental results, you are asked to calculate the local wall heat flux at the location of the measurement.
- Now a change in the mean temperature measurements is performed and this time the temperature is measured using a thermocouple. The thermocouple is placed 5 mm above the wall. What is the expected temperature recorded by the thermocouple?

The aeroplane wing in the direction of the airflow can be assumed as a flat plate for simplicity.

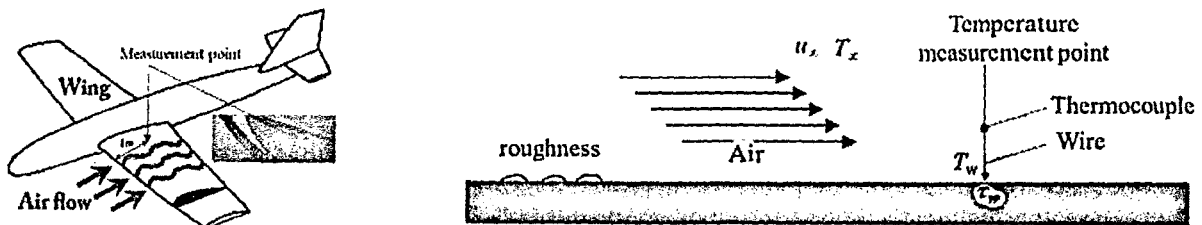


Figure 1. Schematic of the problem

Physical properties of air:

$$\rho = 1\text{ kg m}^{-3}; \nu = 16.7 \times 10^{-6}\text{ m}^2\text{ s}^{-1}; k = 0.025\text{ W m}^{-1}\text{ K}^{-1}; Pr = 0.7; C_p = 10^3\text{ J kg}^{-1}\text{ K}^{-1}$$

Use the temperature distribution obtained in the previous question for the calculation

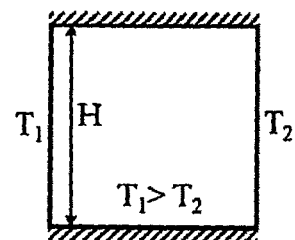
Assume any necessary condition(s) with proper justification(s). Assume the wall coordinate is valid till $y^+ \leq 200$

[2+3=5]

4. A square enclosure is heated from one side and insulated from the top and bottom. What will be the scale of circulating velocity for which the heat transfer will be dominated by conduction? Highlight the Raleigh number of such systems and sketch the probable velocity profile inside the enclosure. The inertia, viscous/friction, and buoyancy forces are scaled with,

$$\frac{\partial^2 v}{\partial x \partial t}, \nu \frac{\partial^3 v}{\partial x^3}, \text{ and } g\beta \frac{\partial T}{\partial x} \text{ respectively.}$$

[2+3=5]



Equations:

➤ Energy conservation equation is given as

$$\rho c_p \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + q''' + \beta T \frac{DP}{Dt} - \nabla \cdot [\tau \cdot \mathbf{v}]$$

The symbols stand for conventional variables and constants.

➤ Momentum balance equation for a Newtonian fluid is given as

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla P + \mu \nabla^2 \mathbf{v} + \mathbf{F}$$

$\mathbf{F}$ is the body force vector per unit volume, and the rest of the symbols are conventional variables and constants.

Part-B

5. Consider the cross-section of a long rectangular bar ($0 < x < a$, $0 < y < b$) made from a material with constant thermo-physical properties. Internal energy is generated at a constant rate Q per unit volume. The surfaces at $(x = 0, y)$ and $(x, y = 0)$ are insulated. A constant heat flux Q_1 leaves the surface at $(x = a, y)$ and the surface at $(x, y = b)$ receives a constant heat flux Q_2 .

(a) Determine the relationship among Q , Q_1 , and Q_2 at steady-state.

(b) Does the steady-state problem have a unique solution for $T(x, y)$? Justify. [2+2 = 4]

6. Consider a long solid cylinder of circular cross-section (Figure - B1). The surface of the cylinder is held at an arbitrary temperature $f(\phi)$. There is no internal energy sources or sinks. Assuming constant thermo-physical properties, obtain an expression for the steady-state temperature distribution $T(r, \phi)$ in cylinder using **Separation of Variables Method**. [6]

Given: In *cylindrical* coordinate system, the Laplacian of temperature T is

$$\nabla^2 T = \frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2}$$

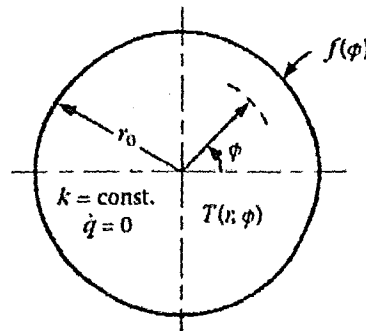


Figure – B1

7. A one-dimensional slab of thickness $2L$ (extending from $x = -L$ to $x = L$) is initially at a uniform temperature T_0 . For times $t \geq 0$, internal energy is generated in the slab at a rate

$$Q = Q_0 \{1 + \beta(T - T_0)\}$$

where Q_0 and β are given constants, while the surfaces at $x = L$ and $x = -L$ are maintained at the initial temperature T_0 . Assuming constant thermo-physical properties, obtain an expression for the unsteady-state temperature distribution $T(x, t)$ in the slab for $t > 0$ using **Finite Fourier Transform Method**. [6]

8. Consider the one-dimensional solid slab of thickness L (Figure B2). It is initially at the fusion temperature T_f . The surface at $x = 0$ is suddenly maintained at a time-dependent temperature given by $T_L(0, t) = T_0 e^{\beta t}$ where T_0 and β are constants. Since $T_L(0, t) > T_f$, a liquid-solid interface forms at $x = 0$ and moves towards the positive direction of x . Use a quasi-steady state model to determine the time it takes for the slab to melt completely. Assume constant thermo-physical properties of the liquid and solid phases. [5]

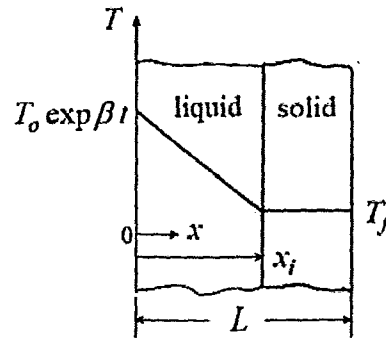


Figure – B2

9. Cryosurgery is a technique to treat tumour where extreme cold is used to freeze and destroy abnormal cells by introducing liquid nitrogen through a cryoprobe into the targeted region. Consider the one-dimensional model representing the freezing of biological tissue of length L during cryosurgery (Figure B3). A cryoprobe with temperature $T_0 = -196^\circ\text{C}$ is applied at $x = 0$, while the other end $x = L$ is maintained at an adiabatic condition. In the frozen region, blood perfusion and metabolic heat generation are zero. The thermo-physical properties for both frozen and unfrozen phases are constant. Assume hyperbolic (non-Fourier) heat conduction in biological tissue.
- (a) Formulate the heat conduction problem for temperature distribution in both the phases.
- (b) Show that the boundary condition representing the location of the interface (written in part (a)) is non-linear. [2+2 = 4]

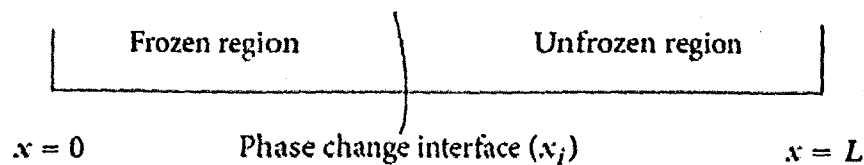


Figure – B3

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